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$$r = \sqrt{(-1)^2 + 1} = \sqrt{2}$$

$$\tan \alpha = \frac{-1}{1} = -1$$

$$\alpha = -\frac{\pi}{4} + \pi = \frac{3\pi}{4}$$

$$\begin{aligned} & \left(\sqrt{2} \operatorname{cis} \left(\frac{3\pi}{4} \right) \right)^{10} \\ &= (\sqrt{2})^{10} \operatorname{cis} \left(10 \cdot \frac{3\pi}{4} \right) \\ &= 32 \operatorname{cis} \left(\frac{15\pi}{2} \right) \\ &= \frac{15\pi}{2} - 2\pi \cdot 3 = \frac{15\pi}{2} - \frac{12\pi}{2} = \frac{3\pi}{2} \\ &= 32 \operatorname{cis} \left(\frac{3\pi}{2} \right) \\ &= 32 \cdot (-i) = -32i \end{aligned}$$

References